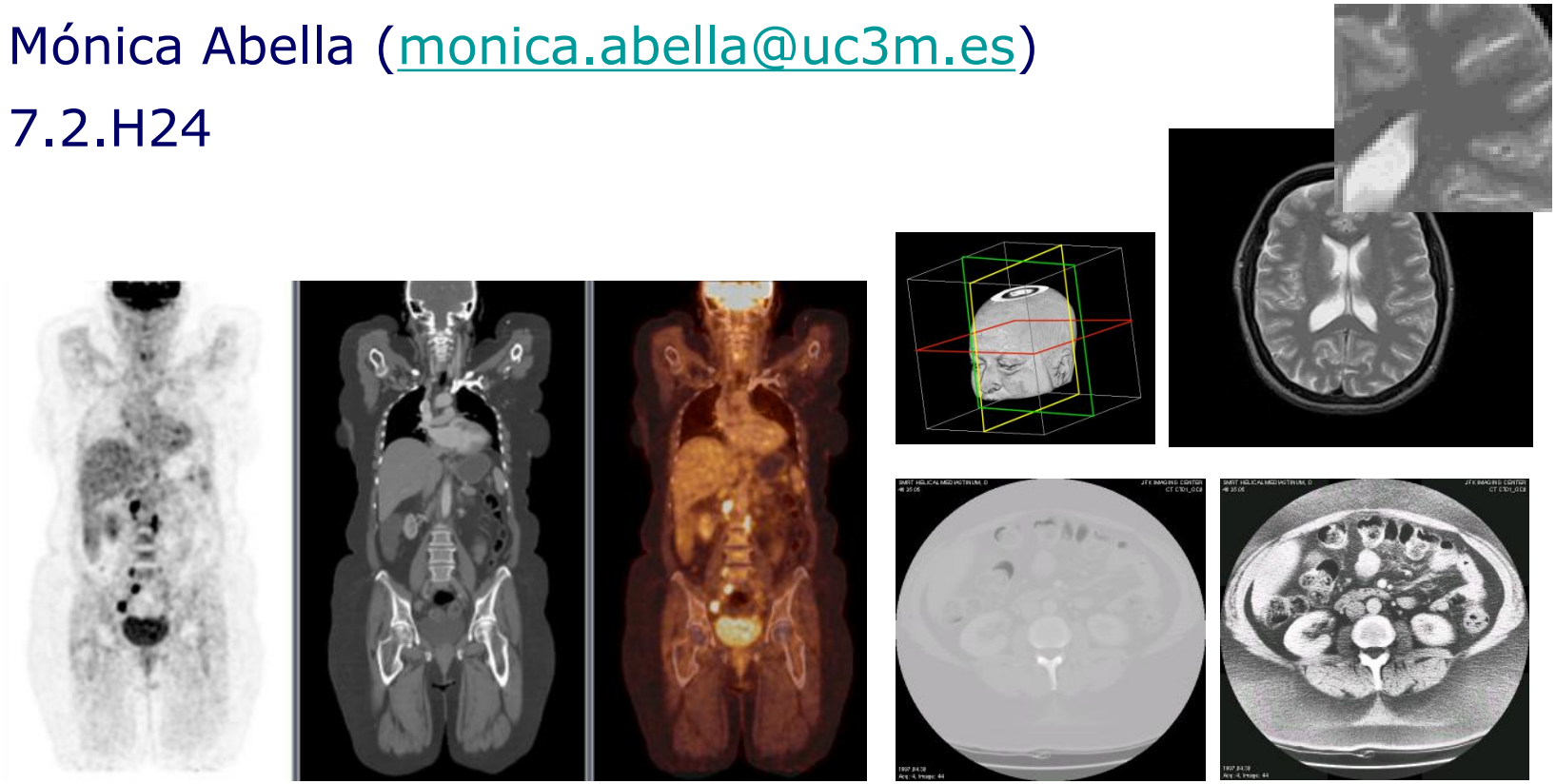
uc3m | Universidad Carlos III de Madrid

Introduction to Bioengineering

Digital Image Basics

Mónica Abella (monica.abella@uc3m.es)

7.2.H24

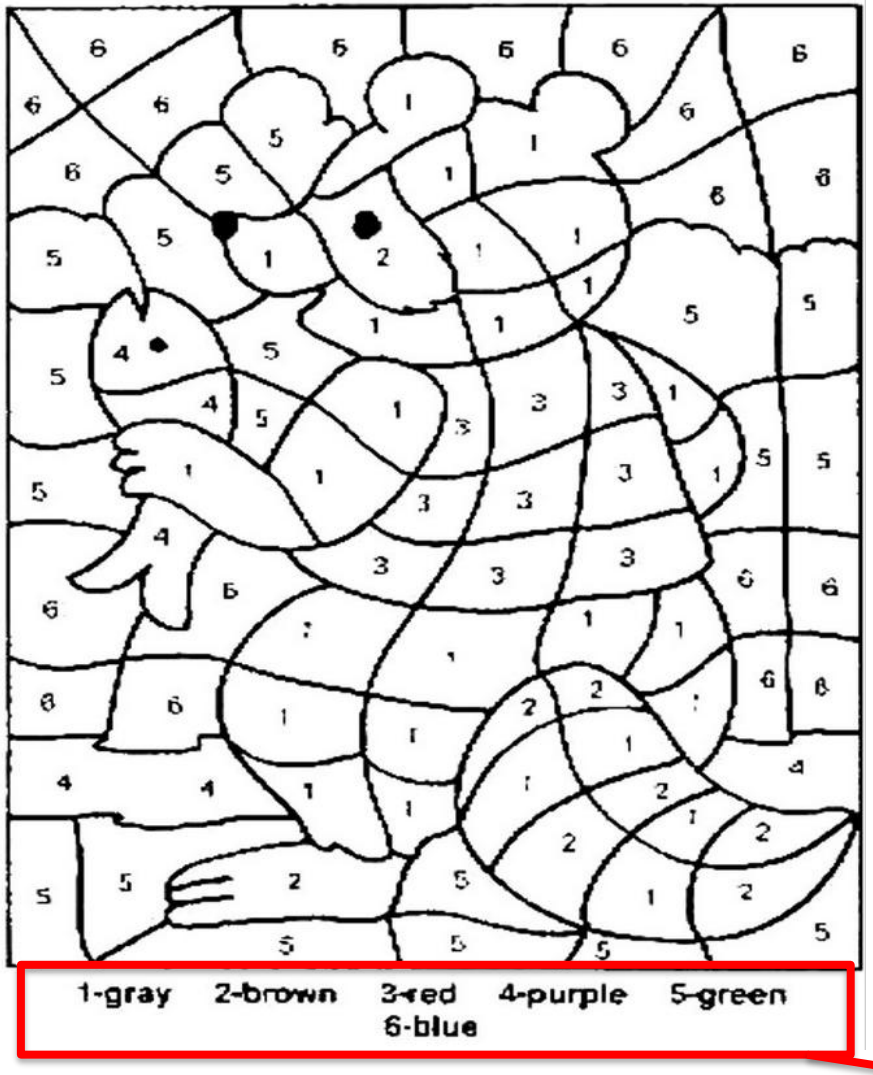


Medical image

- Digital medical images are a representation of the spatial distribution (map) of some physical or chemical property in the organism-> **numbers** not a picture!
- To represent this map as a picture we can assign gray levels or colors to each value

0	1	1	1	0	0	0	0	1	0
0	1	1	1	0	0	0	1	1	0
0	1	1	1	0	0	0	0	1	0
0	1	2	4	0	7	7	2	1	0
0	1	2	5	0	8	8	3	1	0
0	1	2	4	0	8	8	2	1	0
0	1	2	2	0	2	2	2	1	0
0	1	2	5	0	5	5	1	1	0
0	1	2	5	0	2	2	2	1	0
0	1	1	1	0	1	1	1	1	0

Image display





Look-up table (LUT)

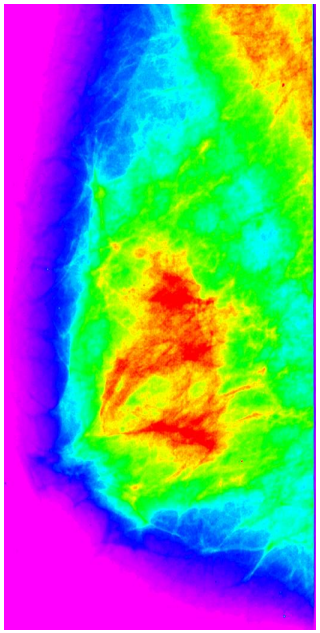
Medical image

- Digital medical images are a representation of the spatial distribution (map) of some physical or chemical property in the organism-> **numbers** not a picture!
- To represent this map as a picture we can assign gray levels or colors to each value

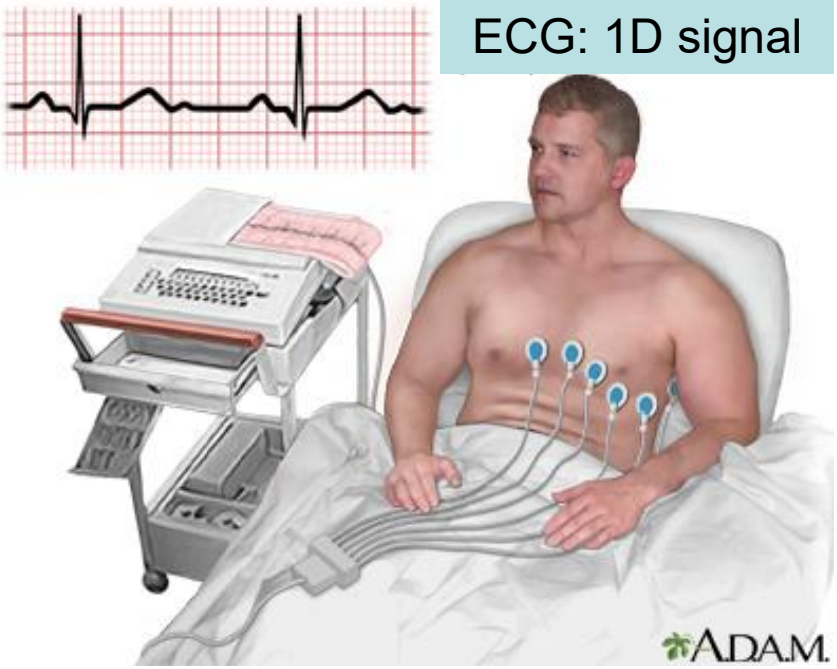
0	1	1	1	0	0	0	0	1	0
0	1	1	1	0	0	0	1	1	0
0	1	1	1	0	0	0	0	1	0
0	1	2	4	0	7	7	2	1	0
0	1	2	5	0	8	8	3	1	0
0	1	2	4	0	8	8	2	1	0
0	1	2	2	0	2	2	2	1	0
0	1	2	5	0	5	5	1	1	0
0	1	2	5	0	2	2	2	1	0
0	1	1	1	0	1	1	1	1	0

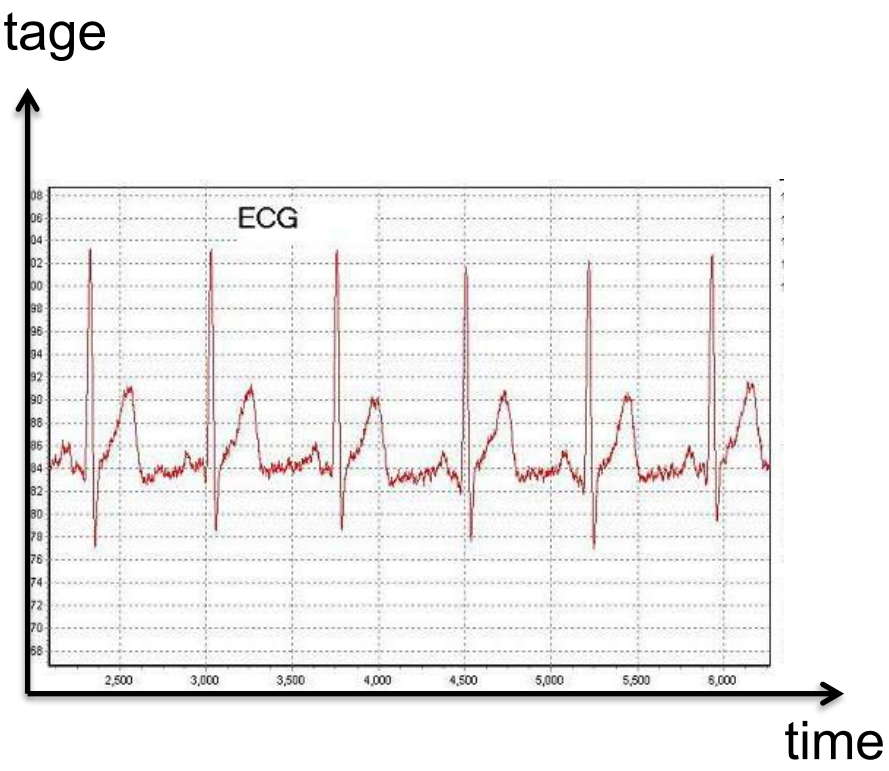
LUT



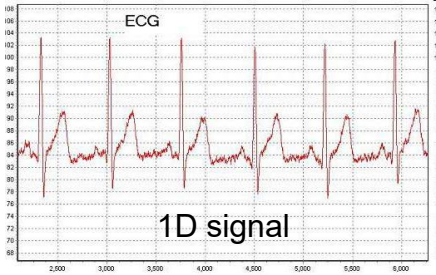


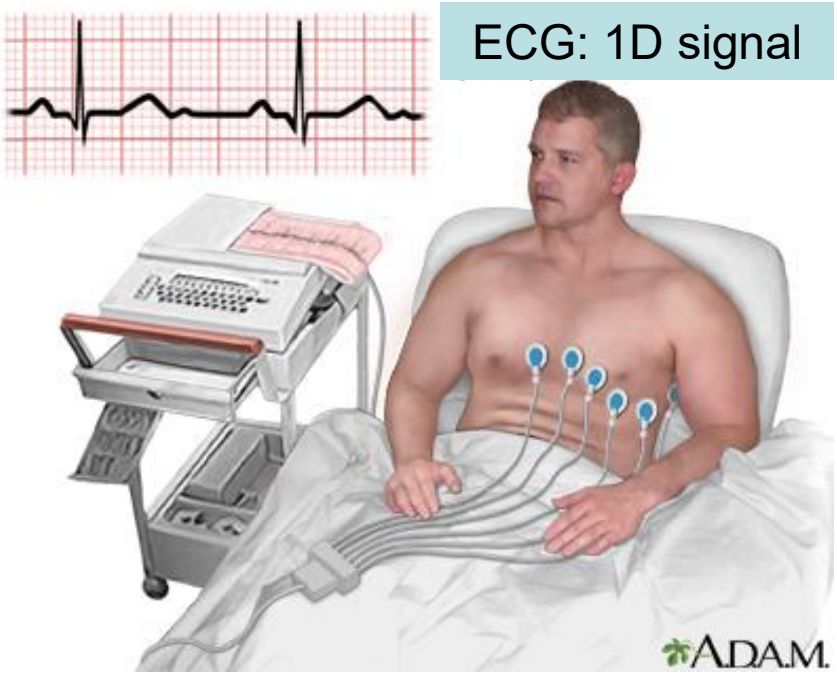
Analog Image: 2D signal

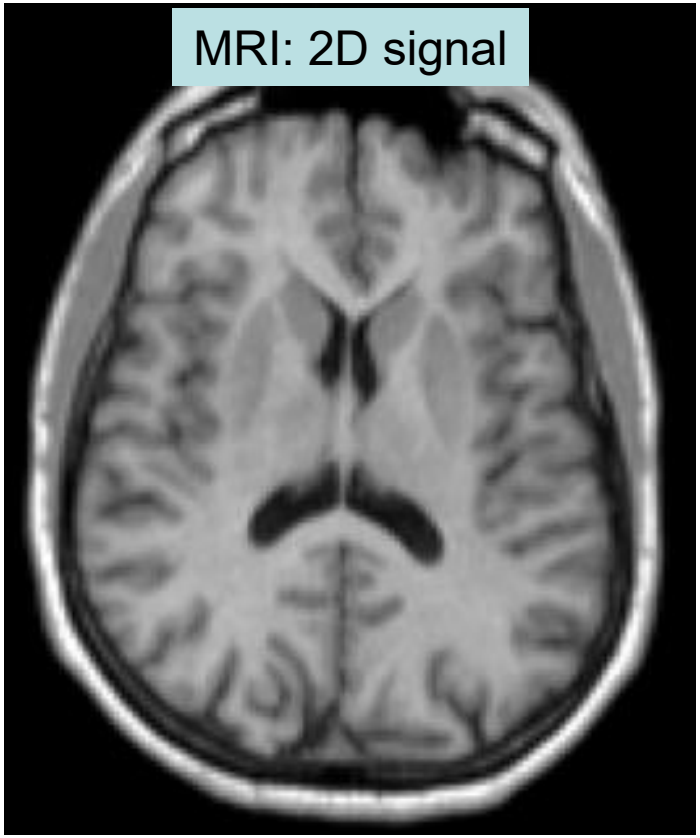




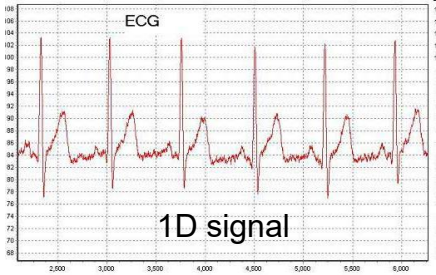
Analog Image: 2D signal

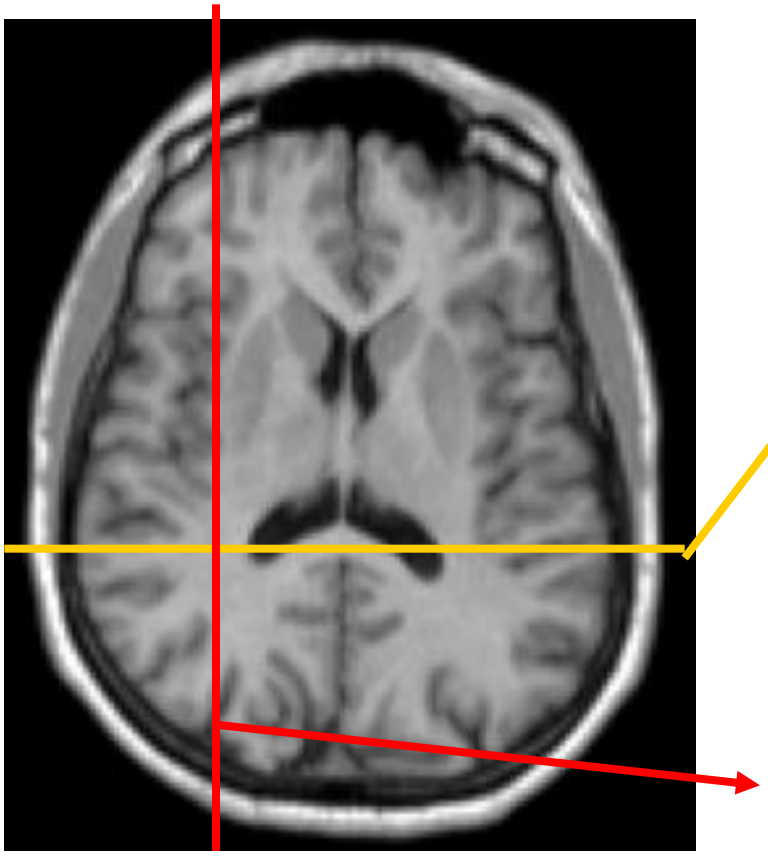


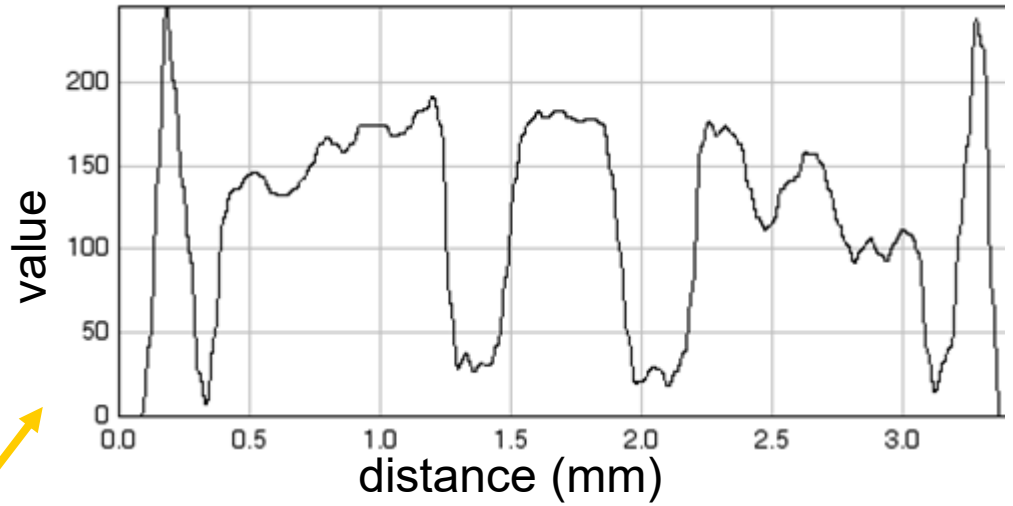


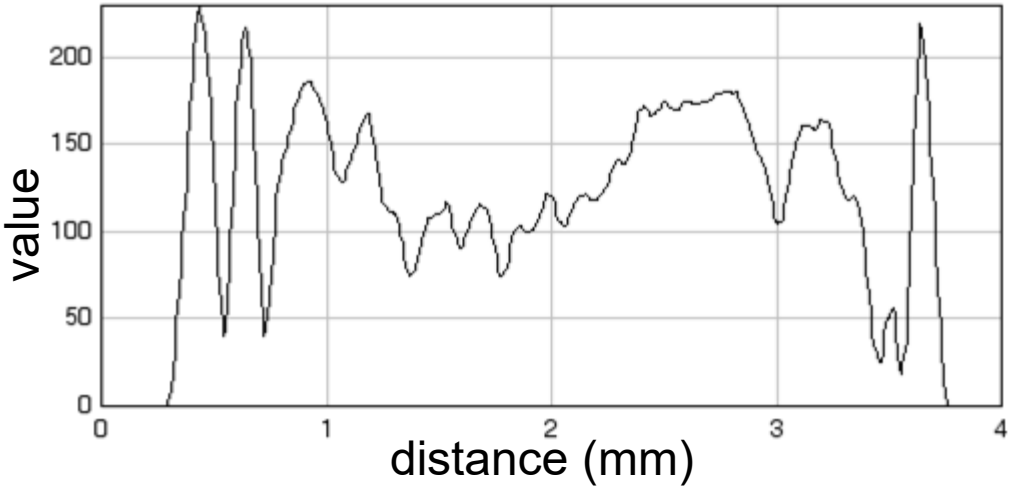


Analog Image: 2D signal

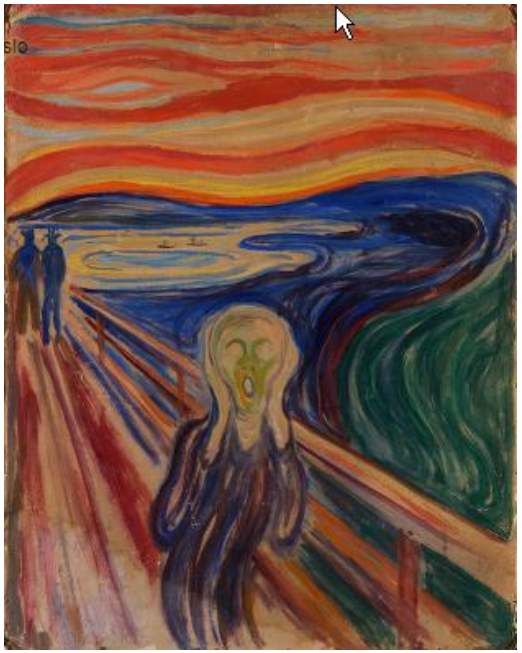






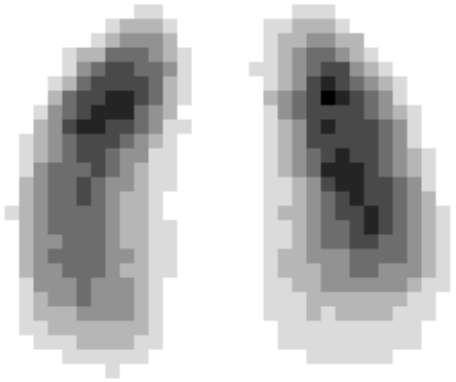


Continuos (analog) – discrete (digital)









Digitization

- Analog signal: There may be any value (**infinite precision**) at any point (**infinite points**)
- To convert an analog 2D signal into a discrete signal
  - when scanning a film radiography
  - when using digital detectors

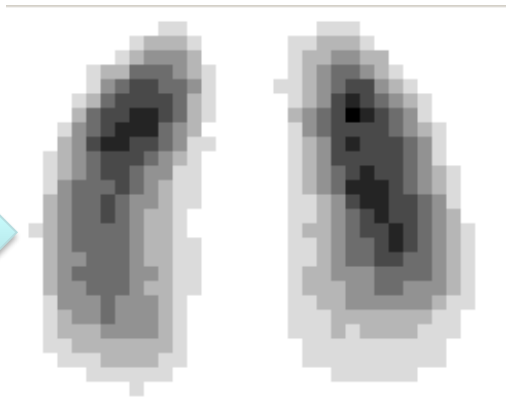




Digitization

0	1	1	1	0	0	0	0	1	0
0	1	1	1	0	0	0	1	1	0
0	1	1	1	0	0	0	0	1	0
0	1	2	4	0	7	7	2	1	0
0	1	2	5	0	8	8	3	1	0
0	1	2	4	0	8	8	2	1	0
0	1	2	2	0	2	2	2	1	0
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0	1	1	1	0	1	1	1	1	0

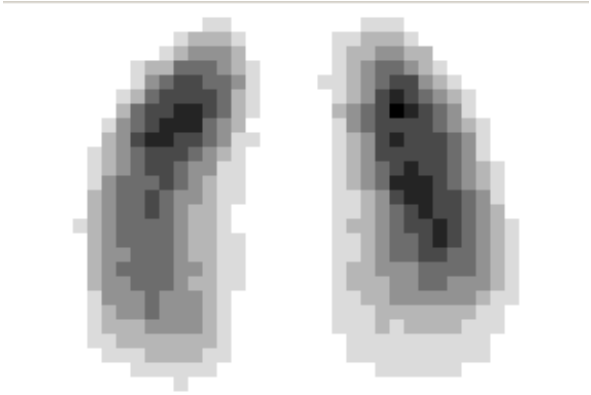
LUT



Digitization

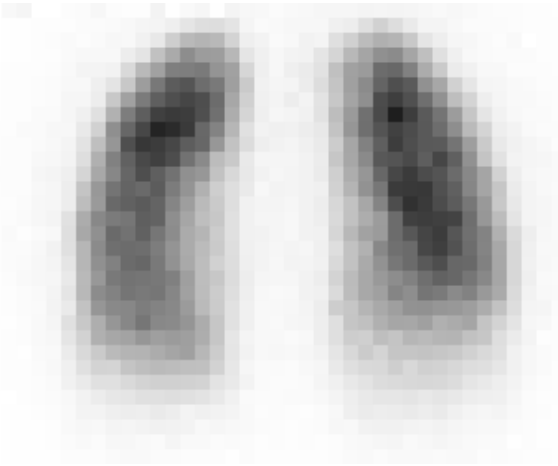


Digitization



Sampling

Quantization



+



Digitization I: Sampling

- Analog 2D signal, *f(x,y)*: infinite points
- We can neither acquire (not ideal acquisition systems) nor store infinite points (finite disk memory)
- We only take a **finite number of samples**

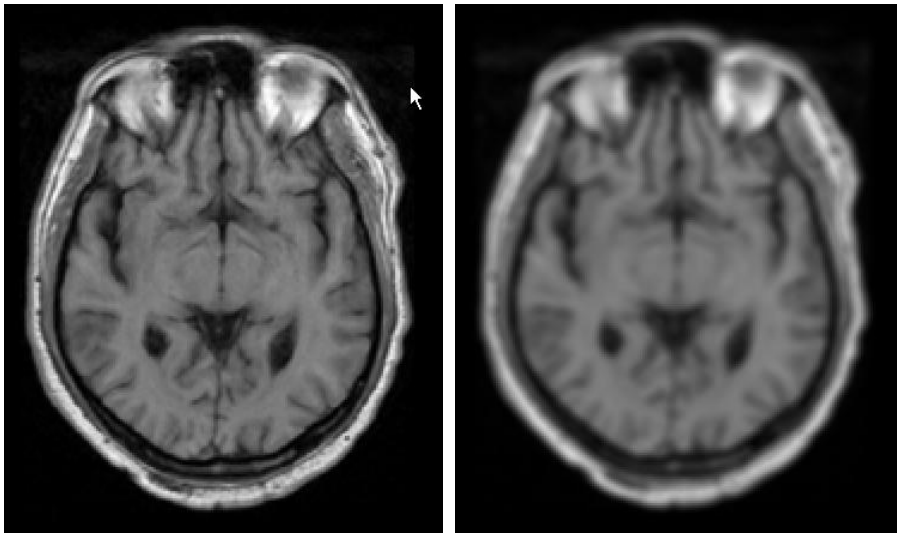
The space is divided in small elements: pixels



Matrix size: number of pixels in each direction

Quality parameters

- **Spatial resolution:**
  Size of the smaller detail we can see in an image



Which image has a higher spatial resolution?



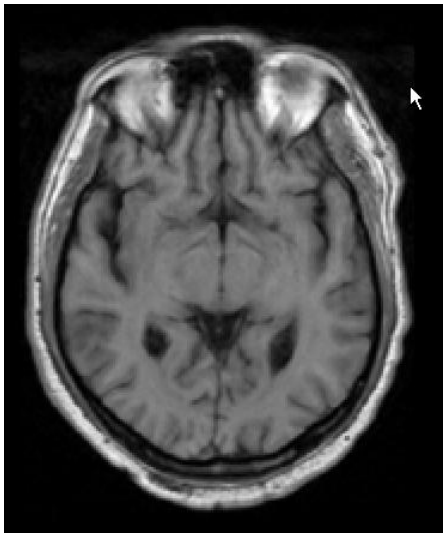
Matriz: 512 × 512

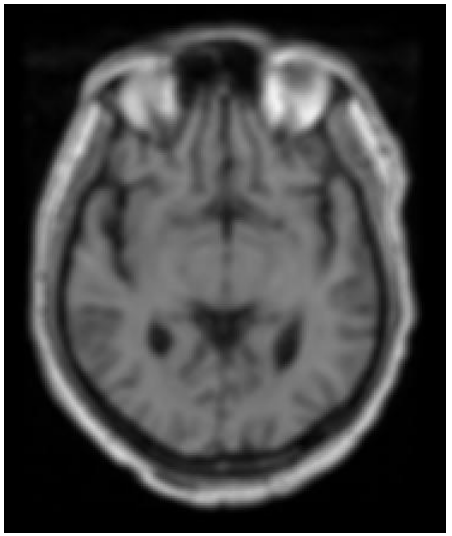


Matriz: 60 × 100

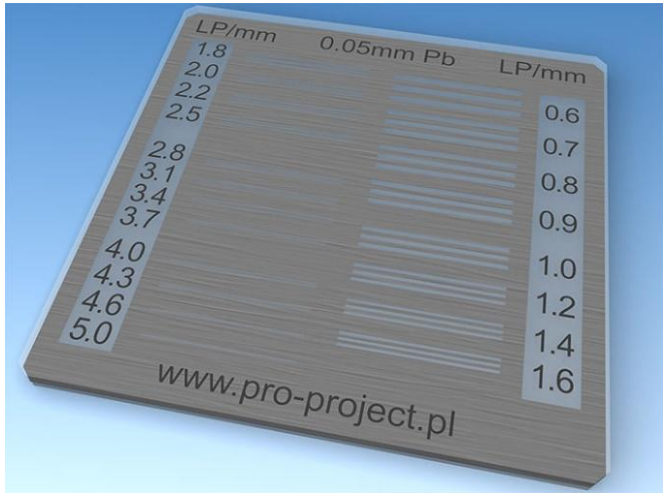
Quality parameters

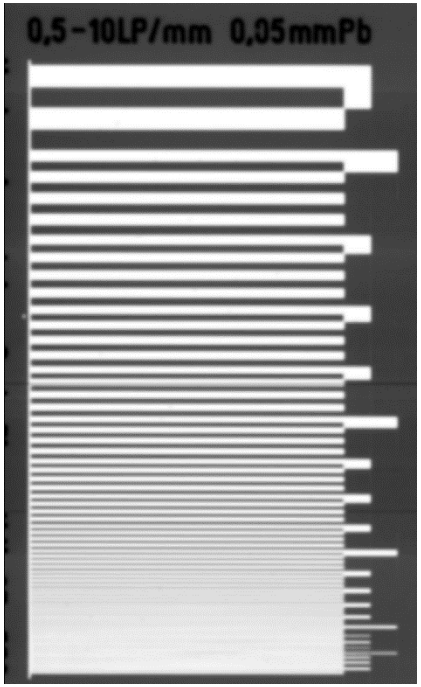
- **Spatial resolution:**
  Size of the smaller detail we can see in an image

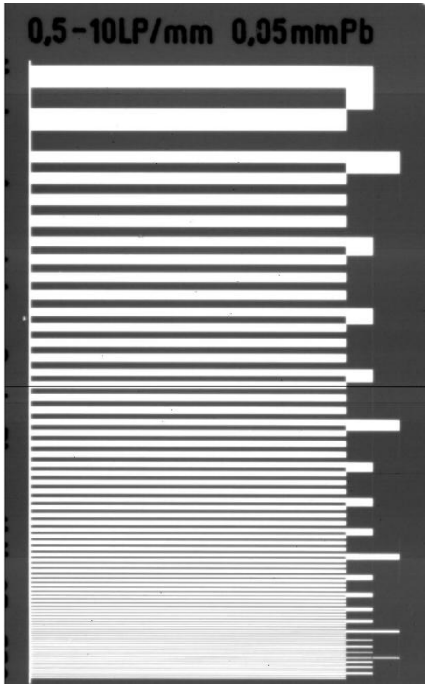




How could I measure the spatial resolution of my imaging system?







Digitization I: Sampling



256 x 256

Different matrix size

What about spatial resolution?



128 x 128

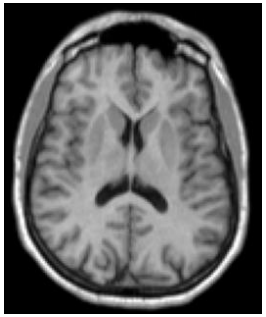


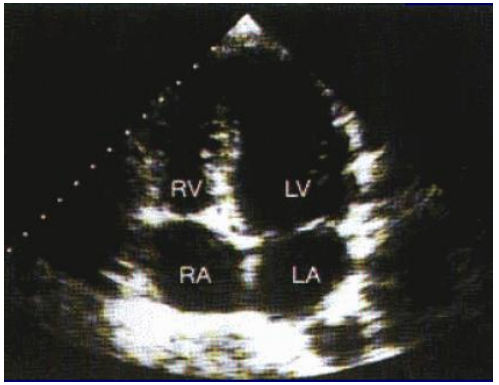
64 x 64



32 x 32

- Field of view (FOV):Real size of the space represented in the image



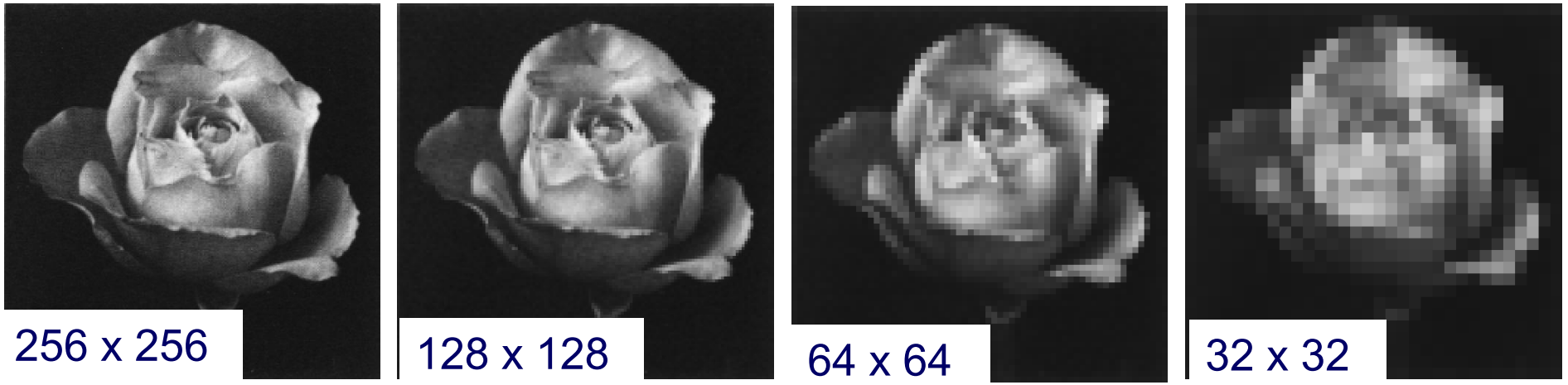


Digitization I: Sampling

Same FOV

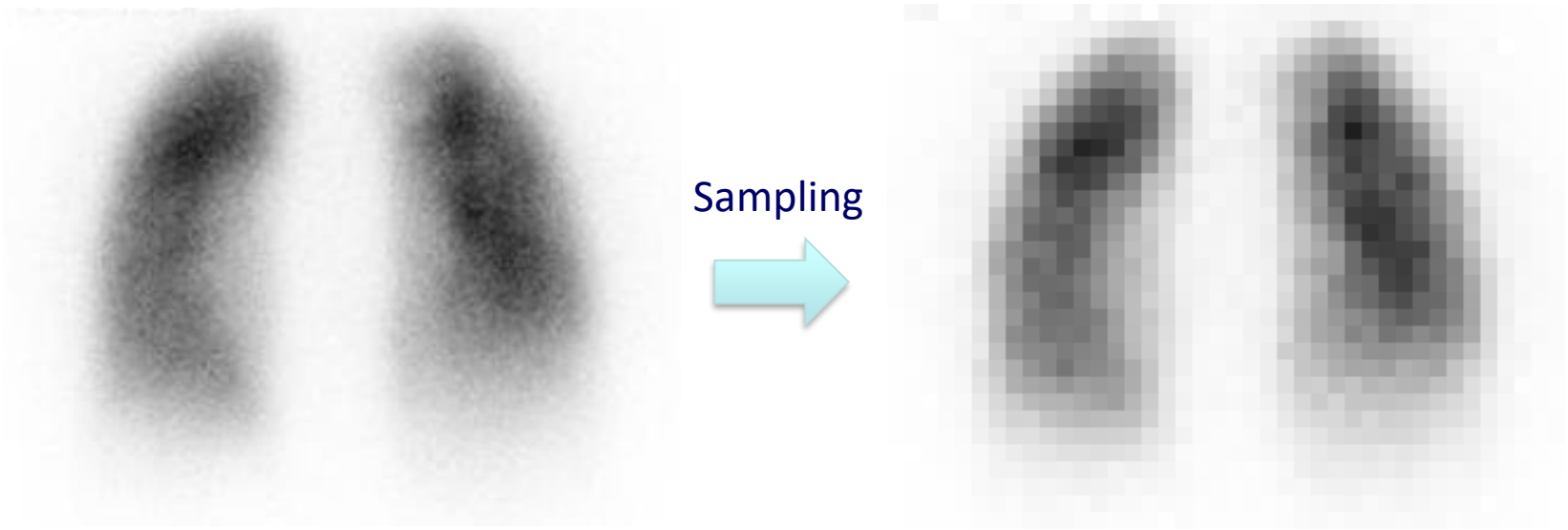
Different matrix size

What about spatial resolution?



Digitalization

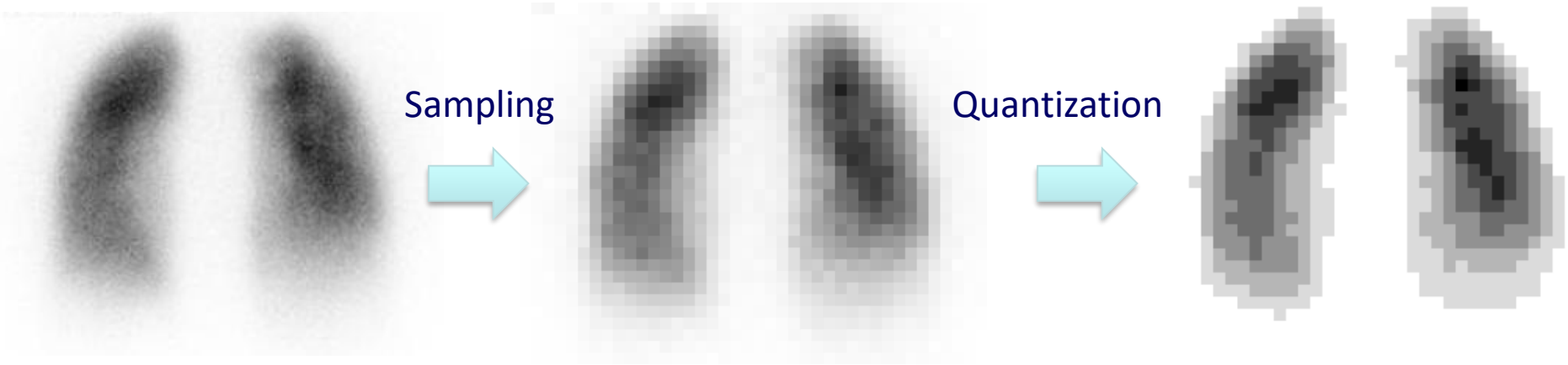
- We have gone from infinite points to a finite number of pixels



Now, what is the value at each pixel?

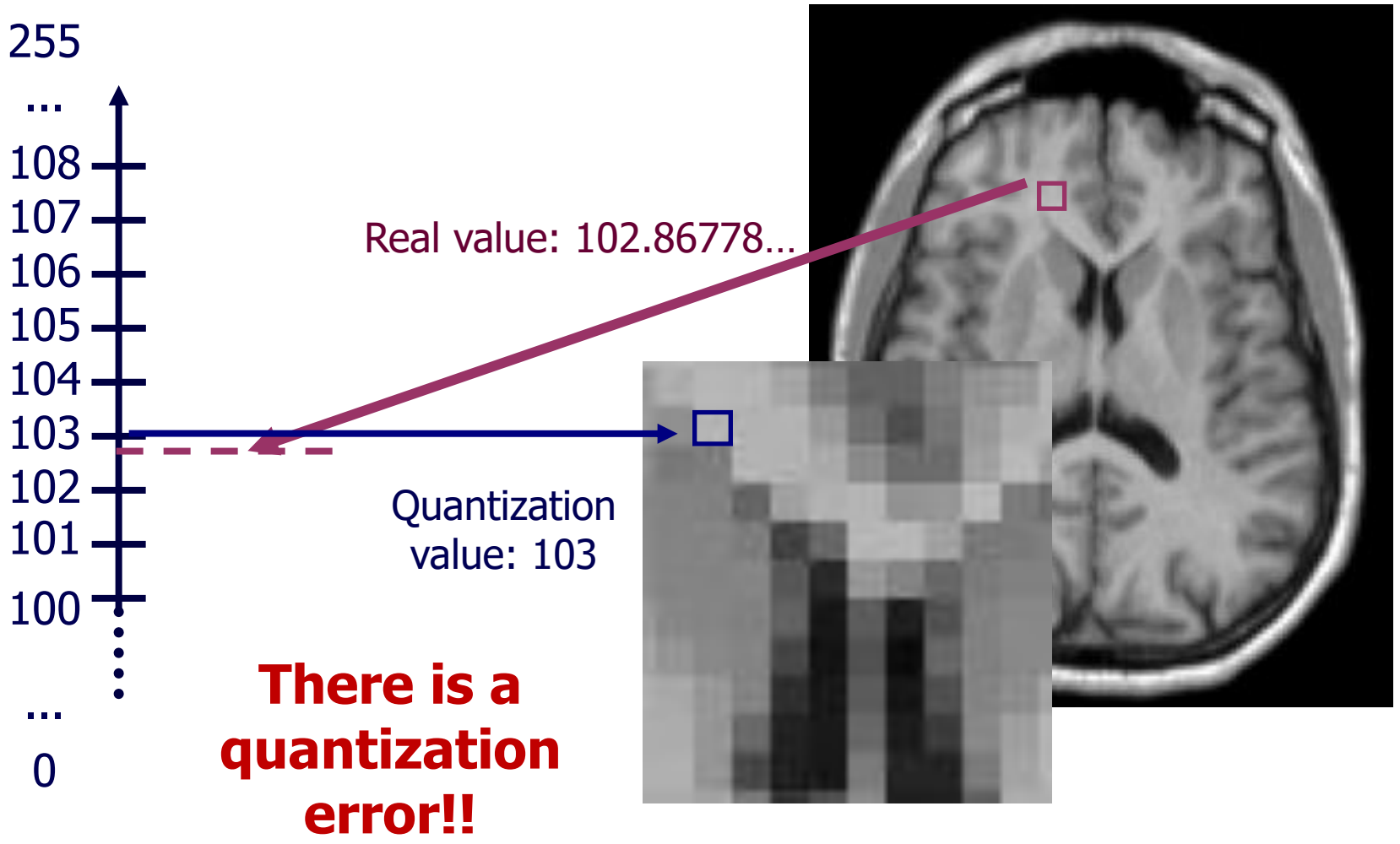
Digitalization II: Quantization

- Points in an analog image can have values arbitrarily precise (it is a continuous distribution)
- We can neither record (limitations in the acquisition) nor store (limitations in disk space) values with infinite precision
- We need to **round** values: The digital images will have a **limited** number of possible values

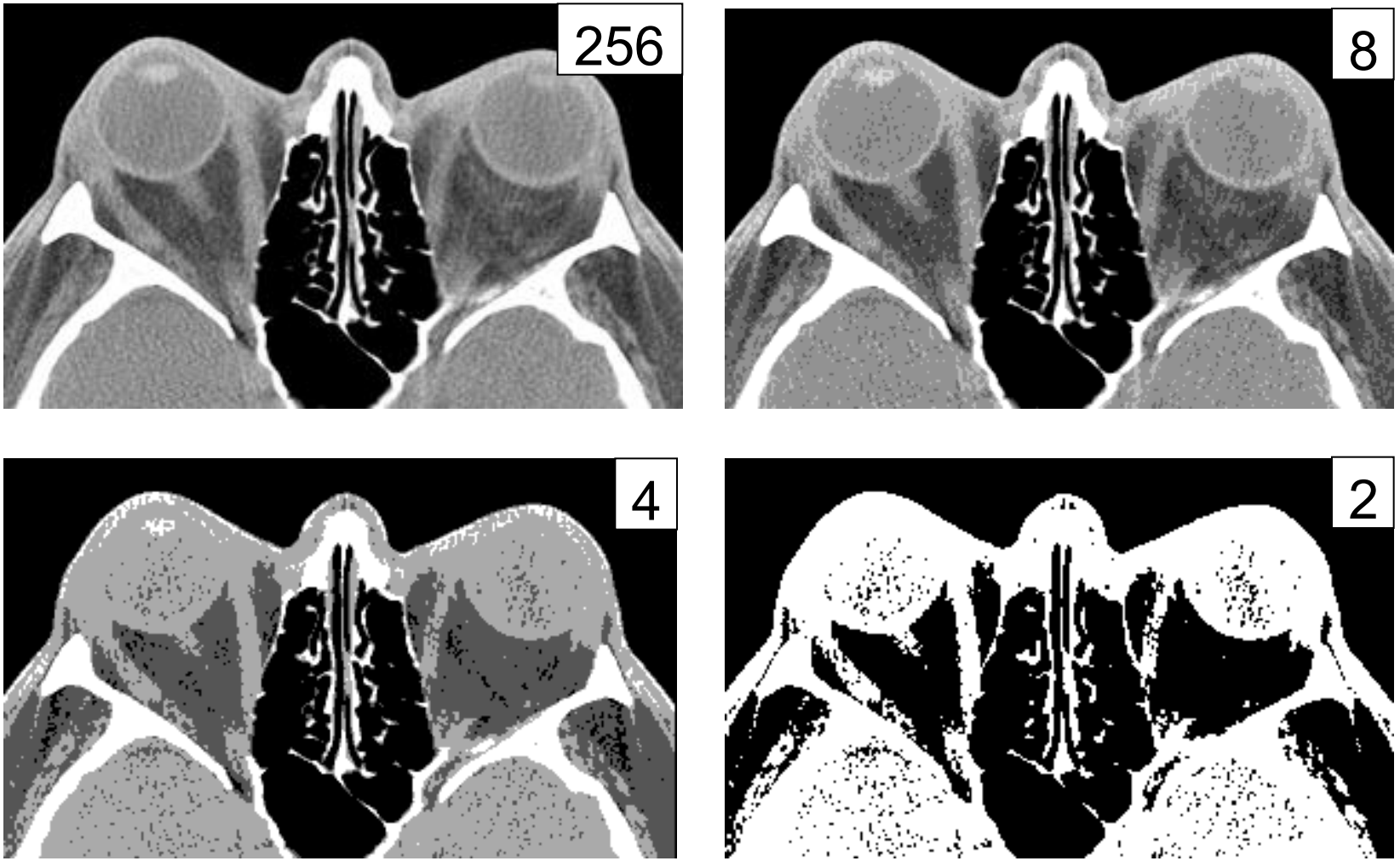


Bit depth: number of bits used for each pixel (defines max. precision achievable)

Digitalization II: Quantization



Digitalization II: Quantization



Digitalization II: Quantization

- False contouring





Digitalization II: Quantization

- Bit depth: number of bits used for each pixel (defines max. precision achievable)
- How many values should I store?
  - Amount of information: how much will I need?
  - Disk space

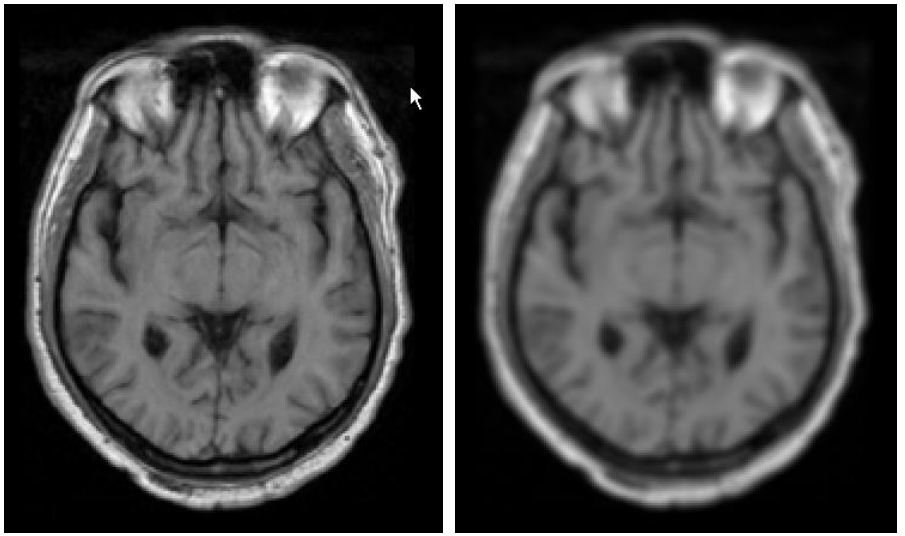
Example: CT

- Hounsfield Units: -1000 .. 3000
  - Its 4000 different values (for HU differences of 1 HU)
  - If I use 8 bits/pixel: 256 different values
  - If I use 12 bits/pixel: 4096 (minimum bit depth for CT)

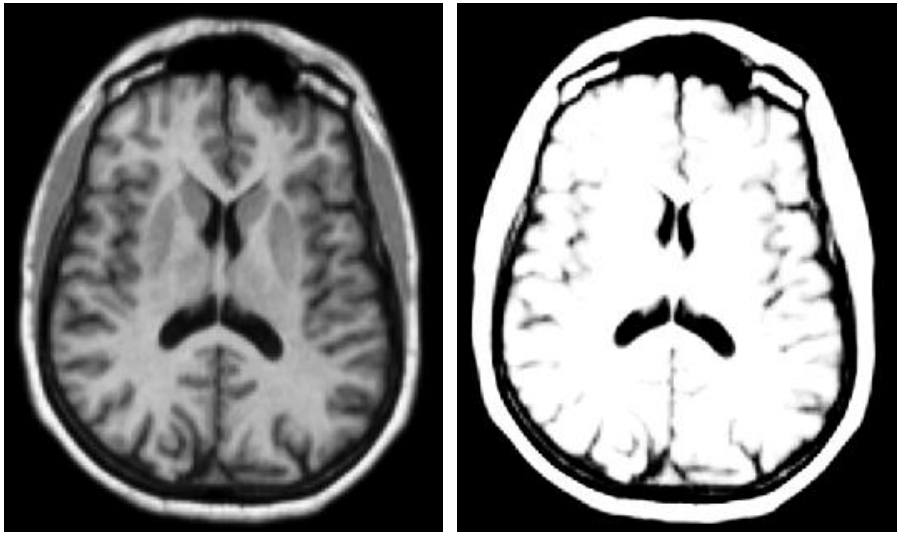


Quality parameters

- **Spatial resolution:**
  Size of the smaller detail we can see in an image

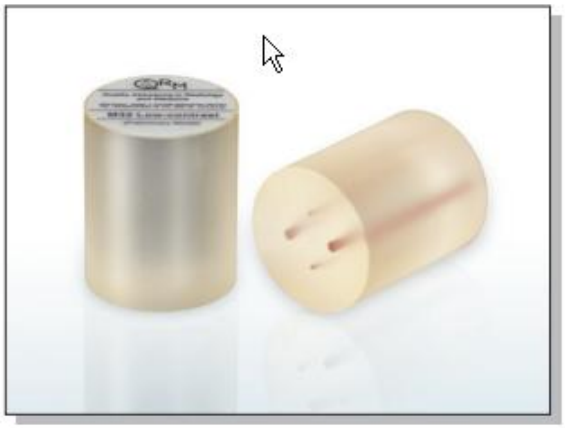


- **Contrast resolution:**
  Ability to distinguish among structures with similar value

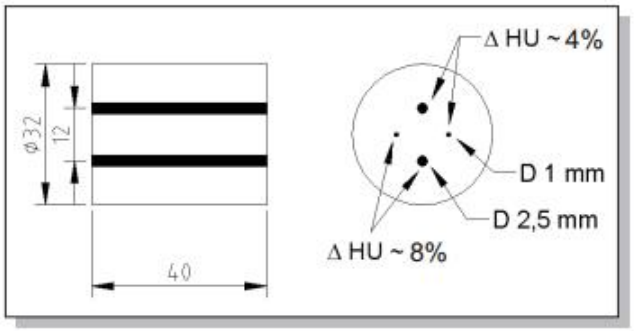


Contrast resolution

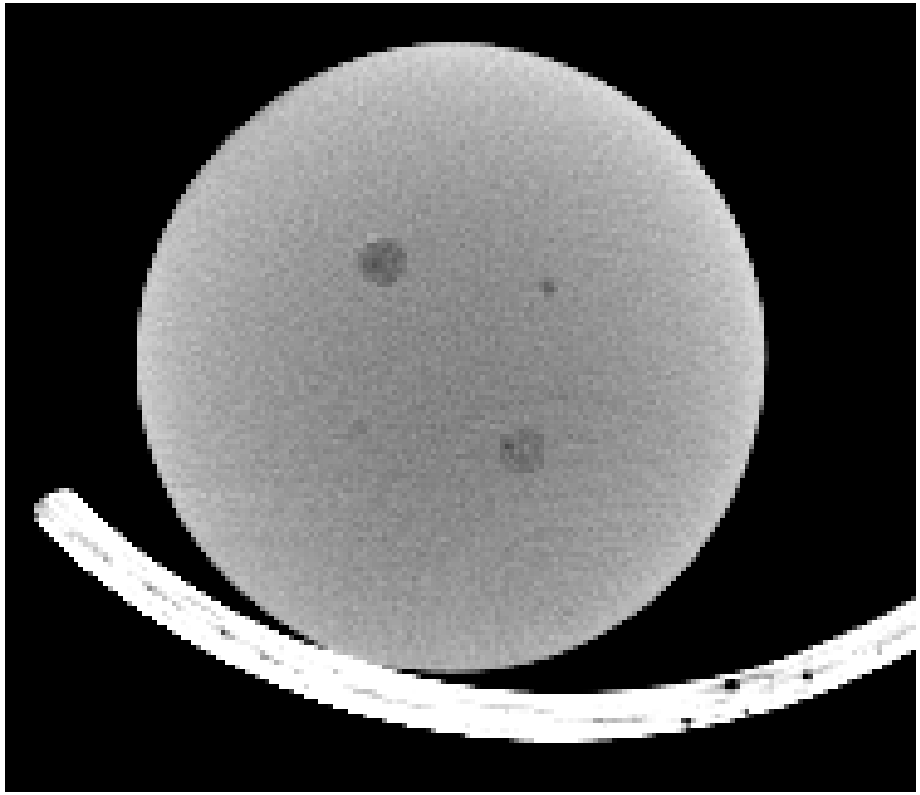
Question: How could I measure the contrast resolution of my imaging system?



Micro-CT Low Contrast Phantom



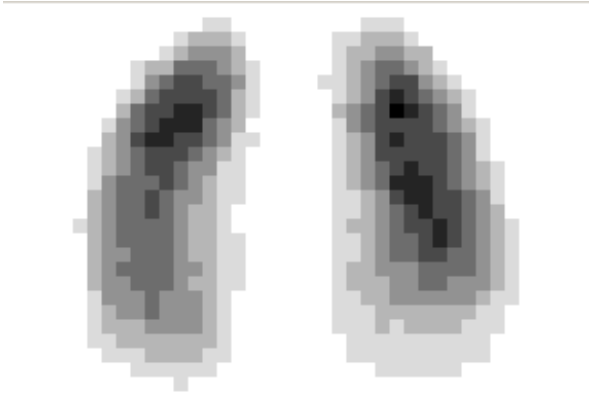
Shematic drawing of the Phantom



Digitization

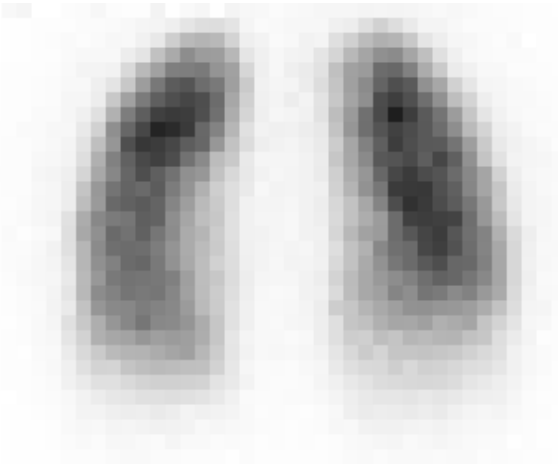


Digitization



Sampling

Quantization

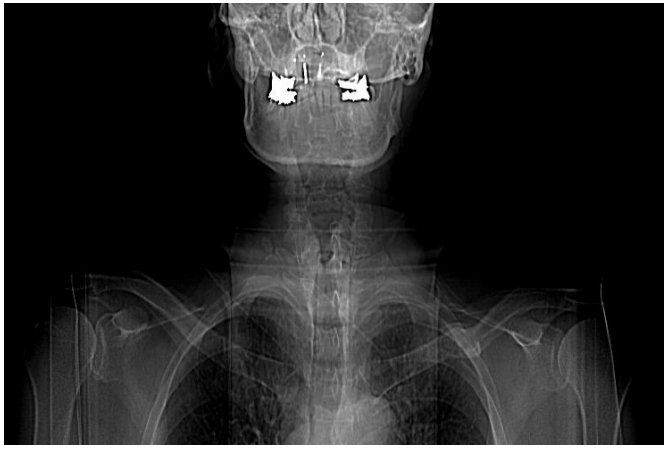


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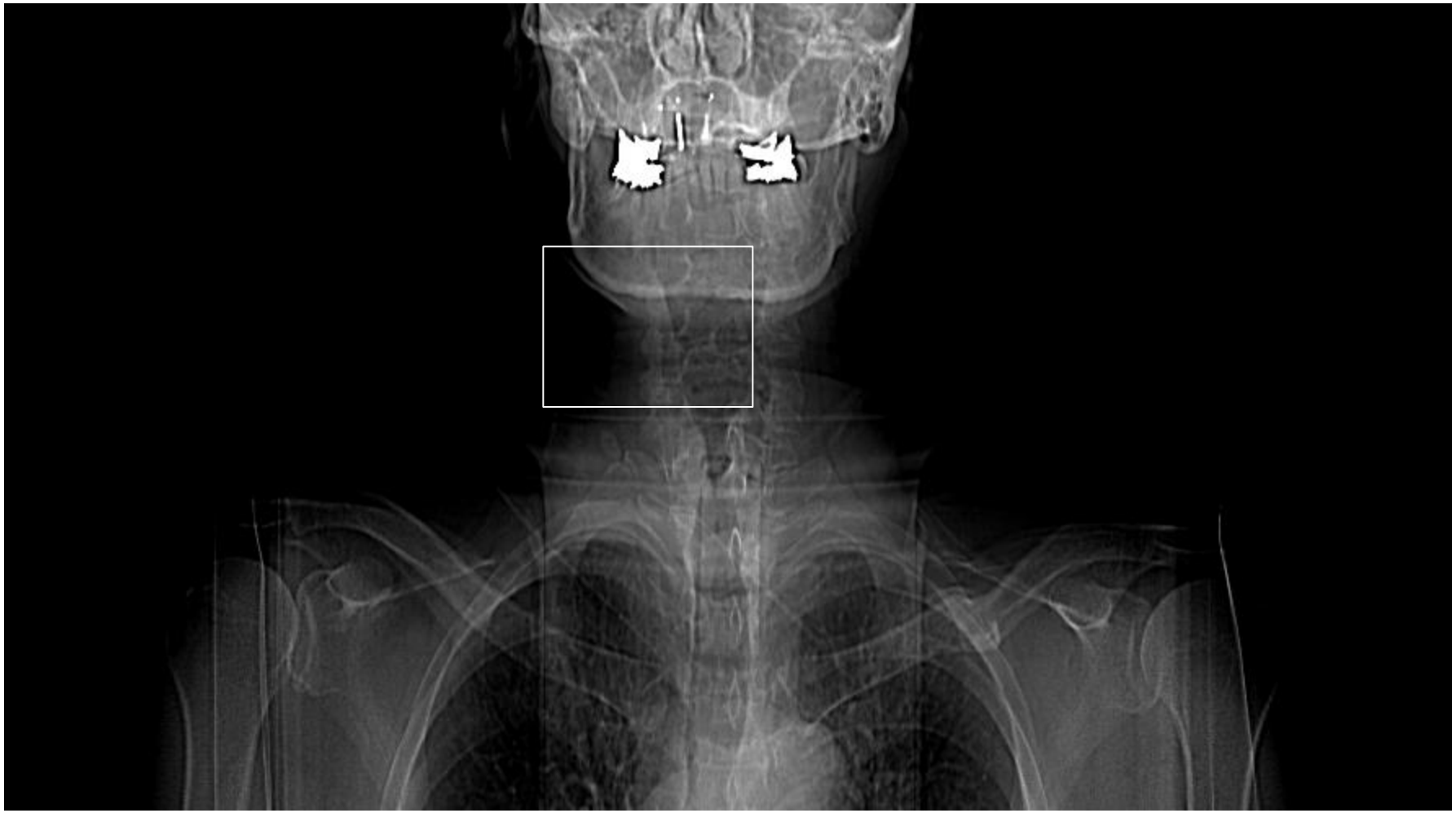


Digital Image

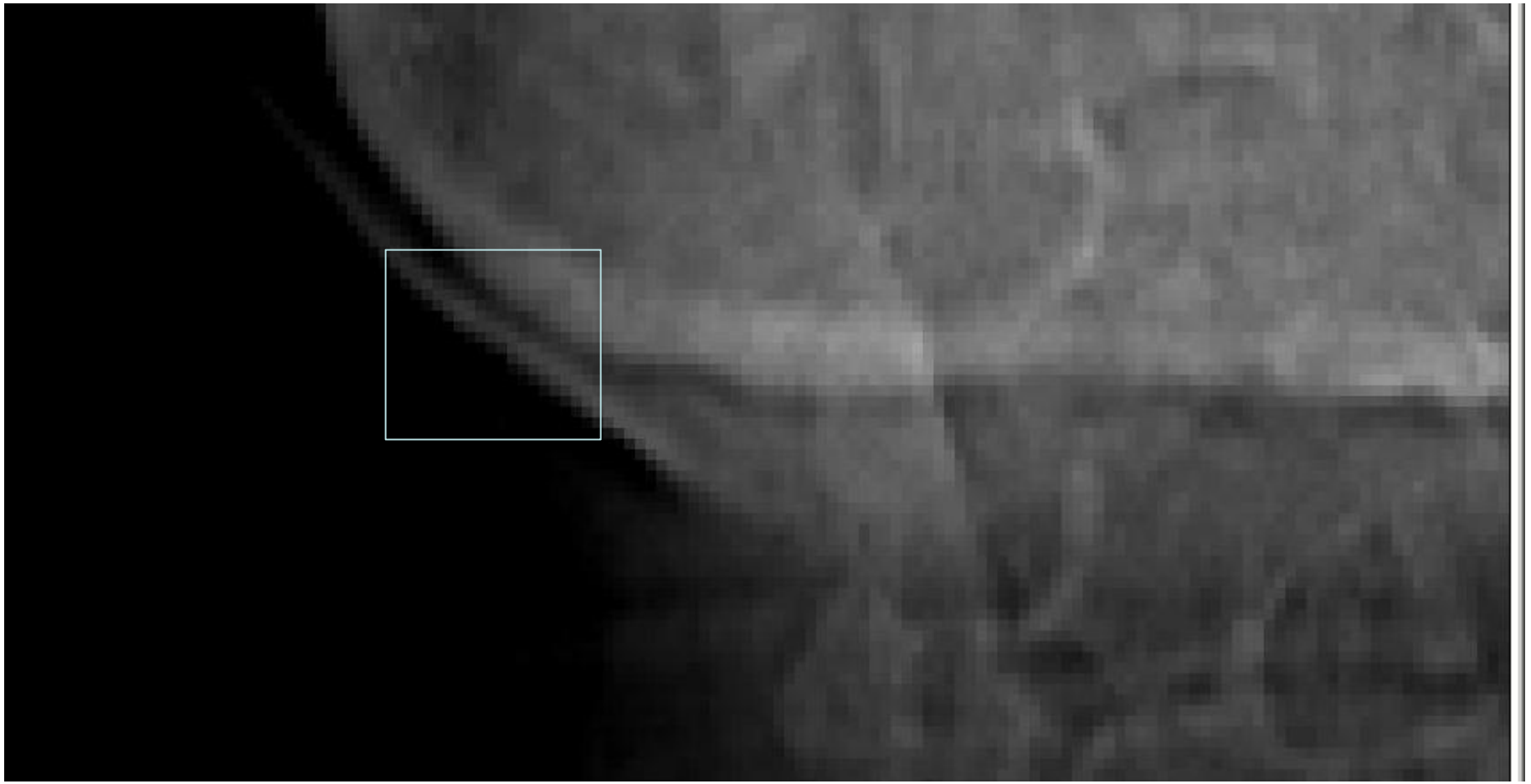
Question: what is a digital image?, what do we actually have stored in the PC?



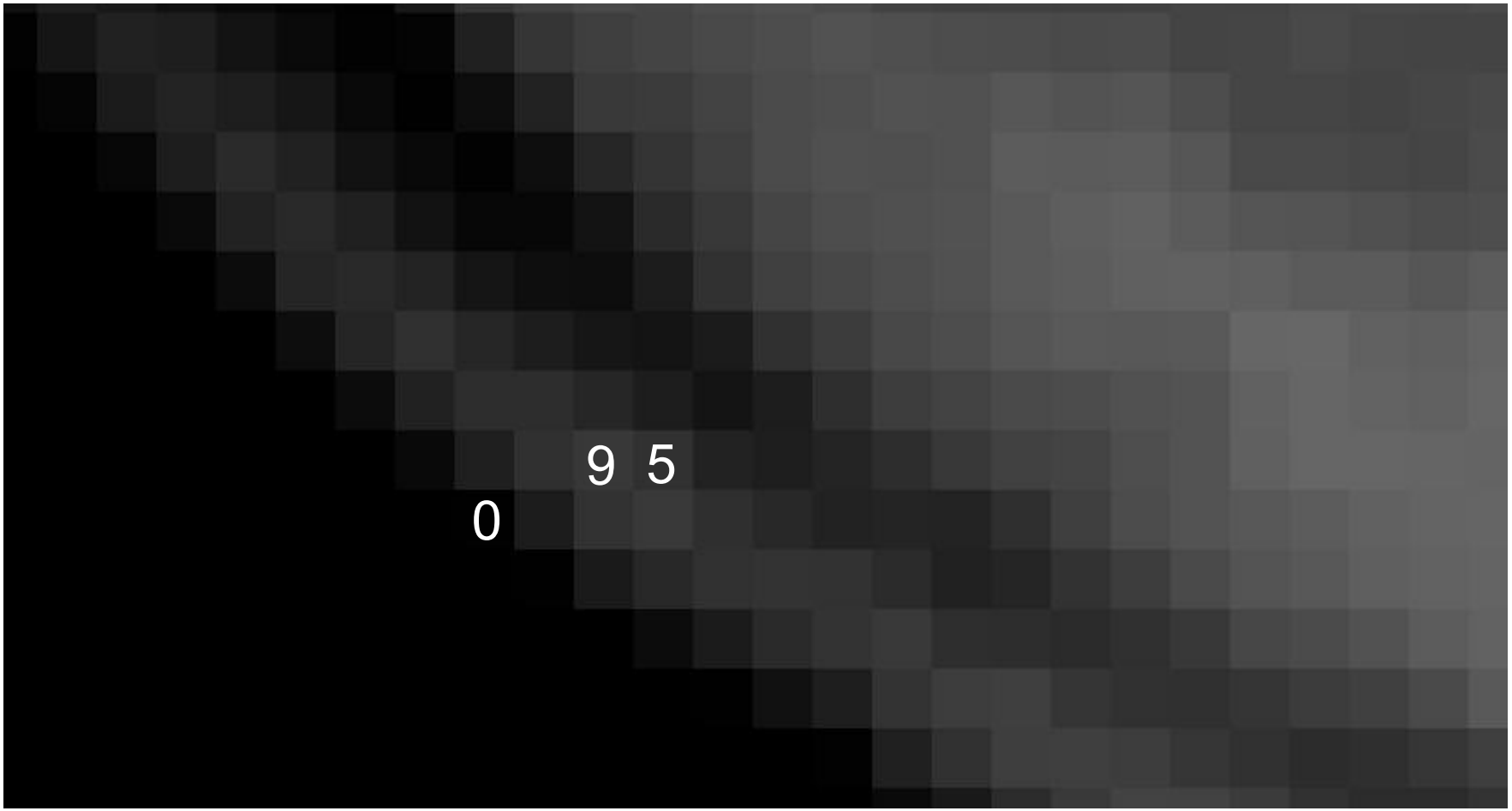
Digital Image



Digital Image



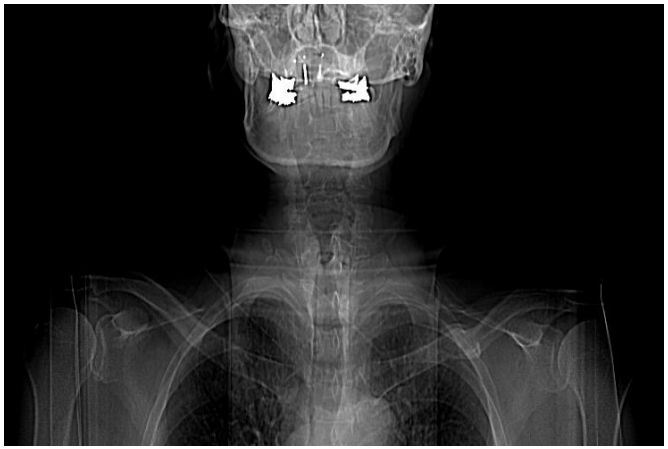
Digital Image



Digital Image

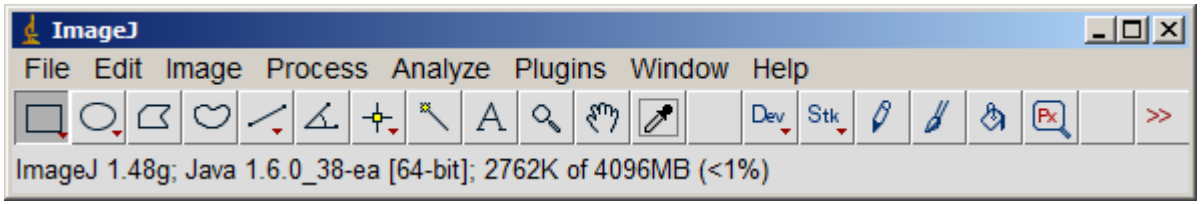
Question: what is a digital image?, what do we actually have stored in the PC?

- List of numbers (really, a list of bits!): value of the image at each point
- Header information: how to arrange those bits? type of data, endianness, size of the image, …
- Different image formats: tiff, jpeg, DICOM, raw,…



ImageJ

http://imagej.nih.gov/ij/



- Java-based image processing program developed at the National Institutes of Health
  - Public domain
  - Designed with an open architecture that provides extensibility via Java plugins and recordable macros

NIH Image to ImageJ: 25 years of image analysis

Caroline A Schneider, Wayne S Rasband & Kevin W Eliceiri

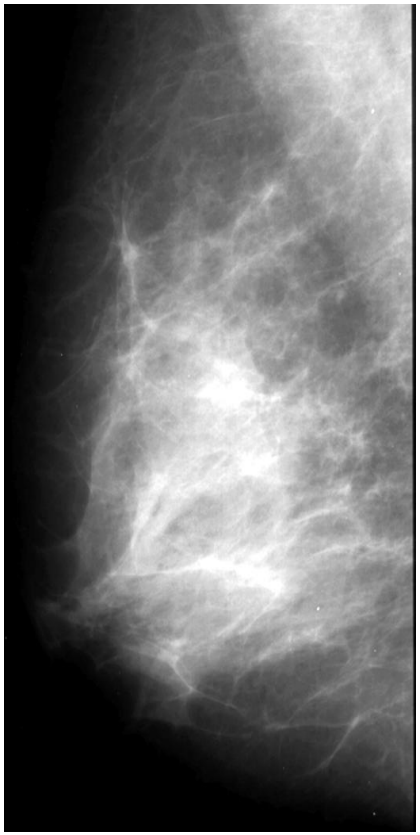
For the past 25 years NIH Image and ImageJ software have been pioneers as open tools for the analysis of scientific images. We discuss the origins, challenges and solutions of these two programs, and how their history can serve to advise and inform other software projects.

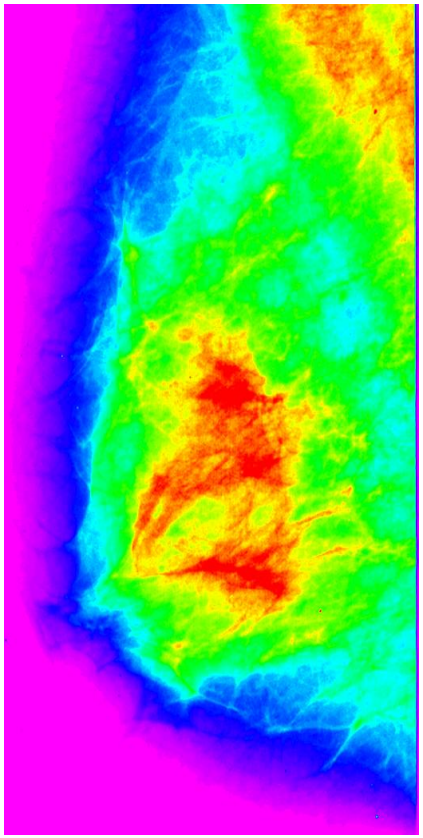
Image formats

- File->Open->Cell Colony
  - Format?
  - Header info?
  - Matrix size?
  - Bit-depth?, number of possible values?
- File->Open->CTslice.dcm
  - Format?
  - Image->Show Info
    - ✓ When was this study acquired?
    - ✓ What is the manufacturer of the scanner?
  - Pixel size?
  - Bit-depth?, number of possible values?
  - Maximum spatial resolution?
  - Can we change the visualization colors? Image->Lookup tables

Pseudo-color

We can not distinguish very similar gray levels





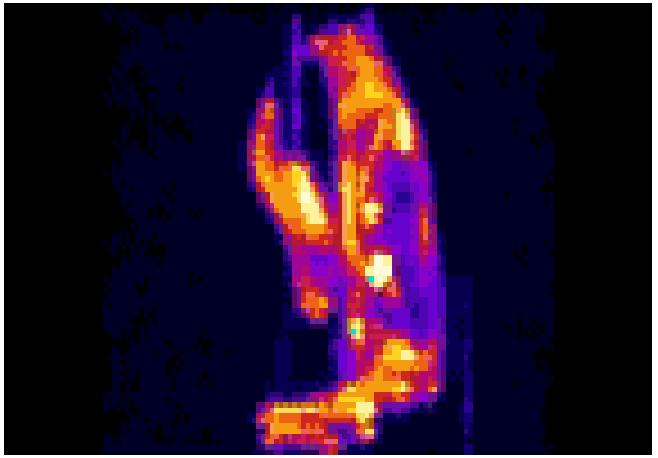


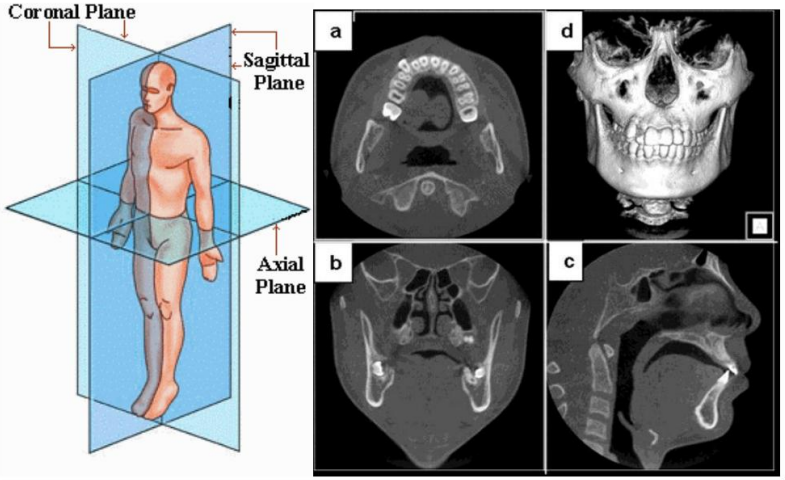


Image formats

- Raw data: Studylopamiro
  - Each study consists of a header file (with extension ".ACT") and a data file (extension ".ctf") stored as: Signed of 16 bits.
  - Try File->Open
  - To open each study we need to used File->Import->Raw:
    1. Open the header file (extension ".ACT") with a text editor (for example use the "Wordpad" application).
    2. Find data type, endianness, and matrix size:
       - Number of pixels on the horizontal direction (Proj_size_radial)
       - Number of pixel son the vertical direction (Proj_size_axial)
       - Number of images (Projections_per_file)
  - Number of possible values?
  - Open the three studies and explain what you see in each one

Image formats and profiles

- File->Open->T1 Head
  - Matrix size?
  - FOV?
  - Bit-depth?
  - Number of possible values?
  - Maximum spatial resolution?
  - *Image->Stacks->Reslice*



- Draw a line in horizontal direction across the image at a height of 190.5 mm using the tool  Plot the profile with Analyze->Plot Profile.
  - What do you see?
  - What do you think these profiles could be useful for?

Resolution-pixel size

- Take just one slice: Image->Duplicate
- With Image->Scale reduce the image size using 0.25 as scale factor in both directions X e Y (X scale, Y scale)
  - Select the option Create new window, to save the current image naming it “Copy” (in Title). Make it bigger with the zooming tool 
    - ✓ What is the field of view?
    - ✓ What is the matrix size?
    - ✓ What is the pixel size?
    - ✓ What is the resolution of the new image?
- Increase the size of image “Copy” with Image->Scale selecting a scale factor of 4 in both directions (X scale, Y scale).
  - What is the pixel size?
  - Have we lost information compared to the original image?
  - What is the resolution of the new image?